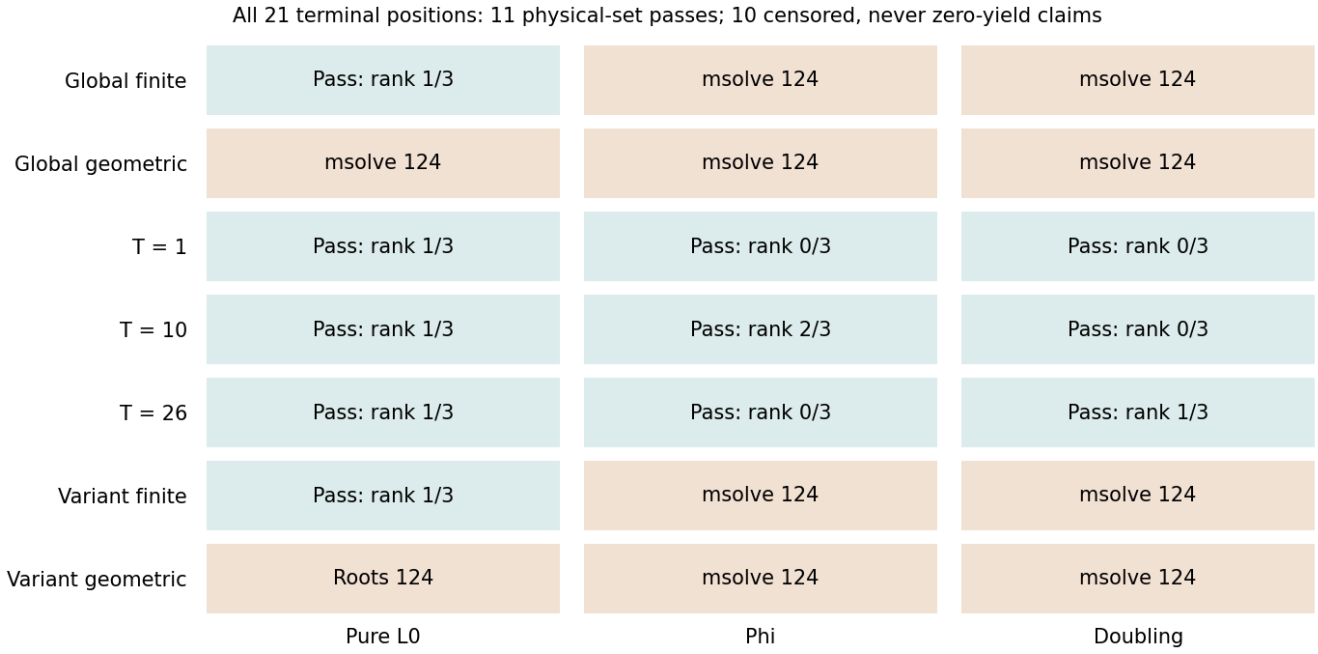
All 21 external S5 positions reconciled with exact physical-set audits

The original 21 msolve processes and both native audit owners are terminal. The corrected follow-up reuses every matching successful or failed receipt and checks only newly available bases. The earlier duplicate retry and its consumed work remain in the separate correction report.



Actual terminal inventory

Every input is the fixed p37/seed0/query2 fixture. P r/3 below means complete physical-set equality with the independent group oracle and independent matrix rank r/3. MS124 is an msolve timeout with no certified basis/root outcome. Root124 is a native root-audit timeout after msolve returned a basis; it supplies no physical-yield result.

Encoding	Pure	Phi	Doubling
Global finite	P 1/3	MS124	MS124
Global geometric	MS124	MS124	MS124
Source block T=1	P 1/3	P 0/3	P 0/3
Source block T=10	P 1/3	P 2/3	P 0/3
Source block T=26	P 1/3	P 0/3	P 1/3
New finite variant	P 1/3	MS124	MS124
New geometric variant	Root124	MS124	MS124

There are 11 physical-set passes, nine external timeouts and one retained native root timeout. All expected positions are present in the workflow inventory. The pure block passes recover the same single physical tuple; their union does not create three relations. The mixed block unions agree with the earlier complete source-block regressions.

Each passing audit verifies input reductions and S-pairs, exhausts the basis's Fp variety, checks allowed factors/signs and group sums, and compares the complete physical set and rank with the separately scoped group oracle. Reverse ideal containment remains unverified. These finite physical-set results do not establish generic equality of ideals.

Scope and receipts

Official msolve 0.11.0, its source/image/executable hashes, one-thread options, exact system/basis hashes and exits are retained in runs/external-regression-v2/. External phases retain 8 GiB, 1,180-second cooperative and 1,200-second outer limits; native audits retain 8 GiB/1,200 seconds and the shared busy reservation. These are not admitted paired solver timings or a complete DLP pipeline comparison. No missing or timed-out result is counted as zero physical yield.

results/external-audit-completion-v2-inventory.json is the unchanged workflow inventory, SHA-256 8c195a6e0c9c8963ddca258e7853941ba37edb2da405b29bb687645668c5a9f3. Native summaries and every original failure remain in the raw archive. No fastest-external, IC speedup or leaderboard promotion is claimed.